

PAPER_723: UQFF Quantum Variable Documents: gamma, E_react, f_quasi, R_b

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Abstract

Five quantum variable documents define: γ (UQFF damping rate for magnetic string oscillations), $E_{react}(t)$ (reaction energy expressing the vacuum energy available for field driving), f_{quasi} (quasi-static field perturbation), R_b (buoyancy Heaviside step radius), and ω_c (UQFF slow-cycle frequency coupling atomic to galactic timescales).

1. Variable Registry

Variable	Value	Physical Role
γ	$5 \times 10^{-5} \text{ day}^{-1}$	Damping in $e^{-\gamma t \cos(\pi t_n)}$
$E_{react}(0)$	10^{46} J	Vacuum energy at UQFF scale
f_{quasi}	0.01	Perturbation: $(1 + f_{quasi})$ in U_m, U_H
R_b	$3.086 \times 10^{19} \text{ m (1 kpc)}$	Heaviside step for U_g2 profile
ω_c	$1.585 \times 10^{-8} \text{ rad/s}$	Slow-cycle: $\mu_j(t) \sin(\omega_c t)$

2. Reaction Energy Decay

$$E_{react}(t) = 10^{46} \cdot e^{-\kappa t} \quad (\kappa = 5 \times 10^{-4} \text{ day}^{-1})$$

At 100 years: $E_{react}(36500) \approx 1.61 \times 10^{44} \text{ J}$.

3. Buoyancy Step Profile

$$U_{g2}(r) = k_2 \cdot \frac{(\rho_{UA} + \rho_{SCm})M_s}{r^2} \cdot S(r - R_b) \cdot (1 + \delta_{sw} v_{sw}) \cdot H_{SCm} \cdot E_{react}$$

where $S(r - R_b)$ is the Heaviside step activating at $r = R_b = 1 \text{ kpc}$.

References

- UQFF KB grok_share_ba508f76c8e.txt entry #73
- Session 176, v5.33

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